Sean Luka Girgis

Application Support Engineer | SQL, Python/Perl, Production Troubleshooting & Legacy Systems

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior application support, production troubleshooting, and enterprise technology professional with 20+ years of experience across production support, defect investigation, SQL, Python/Perl scripting, Linux/Unix, Windows concepts, deployments, technical documentation, application monitoring, performance analysis, and legacy enterprise systems. Strong background supporting mission-critical applications, building support automation, investigating production incidents, improving uptime and reporting accuracy, and collaborating across development, infrastructure, QA, and business teams. Best aligned to application support roles requiring troubleshooting, scripting, SQL, defect resolution, support tooling, documentation, deployments, and legacy modernization; Java, NodeJS/React, .NET, MongoDB, media asset management, Aspera/Signiant, and modern full-stack ownership are positioned carefully as adjacent or prior/working-knowledge areas rather than overstated current production ownership.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported production reporting, monitoring, and data reliability workflows across large-scale banking infrastructure using BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, Oracle, SQL, Python, and AWS-backed reporting layers.
- Troubleshot production issues by tracing source feeds, application telemetry, SQL behavior, Python processing, failed outputs, missing or late data, latency symptoms, and downstream reporting discrepancies.
- Built Python/Pandas/PySpark and SQL workflows that transformed telemetry from 6,000+ infrastructure endpoints into validated dashboards, reports, forecasting datasets, and operational insights.
- Developed validation checks, reconciliation logic, and regression-style comparisons to identify defects, abnormal trends, unmatched records, and data quality issues before they impacted stakeholders.
- Created and maintained runbooks, metric definitions, data flow notes, support procedures, troubleshooting notes, and stakeholder documentation to improve repeatable support.
- Partnered with application, infrastructure, analytics, engineering, and business teams to prioritize incidents, explain findings, coordinate fixes, and improve operational reliability.
- Used Python, SQL, Unix/Linux-style scripting patterns, Git concepts, and CI/CD-aware delivery discipline to support repeatable validation, reporting, and production troubleshooting workflows.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Provided application performance and production visibility support for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.
- Investigated production and pre-production performance issues by correlating application monitoring data, user-impacting symptoms, and load-test behavior.

- Created dashboards and reporting views that converted monitoring signals into actionable performance, reliability, and stakeholder insights.
- Collaborated with application and infrastructure teams to validate findings, document issues, and support practical performance improvements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance data, replacing manual support/reporting workflows with repeatable processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, SQL/JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during testing and production-readiness reviews.
- Documented test results, expected behavior, defects, risks, and remediation recommendations for engineering and release teams.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led validation and production-readiness support for a high-throughput shopping engine migration from 200+ MySQL nodes to a 6-node Oracle RAC cluster.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized SQL and Oracle access patterns for high-volume workloads while reducing physical hardware footprint by 95%.
- Supported phased cutover, troubleshooting, documentation, and defect resolution for a mission-critical enterprise migration.

Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC) (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, and DB2-backed components in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported billing, customer-management, and financial-style operational interfaces using Oracle, PL/SQL, C/C++, Pro*C, SQL scripts, XML, Java/J2EE concepts, and enterprise messaging patterns.

- Built automation scripts in KornShell/ksh, Perl, Awk, Syncsort, and SQL to support data extraction, transformation, reporting, deployments, validation, and production housekeeping.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, and release support for telecom billing and customer-management applications.
- Maintained code in CVS and ClearCase and supported legacy application environments across HPUX, SunOS, Linux, WebLogic, WebSphere, Oracle, and messaging interfaces.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

Enterprise APM Reporting Automation
Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance telemetry across large financial-services monitoring environments.
Technologies: CA APM, KornShell/ksh, Perl, SQL, Unix/Linux, Automation

Production Data Quality and RCA Workflow
Built validation and reconciliation workflows for production reporting datasets, including row-count checks, null checks, missing feed diagnosis, unmatched join checks, prior-run comparisons, and troubleshooting documentation.
Technologies: Python, SQL, Pandas, Oracle, Data Validation, RCA, Runbooks

Application Performance Monitoring and Support
Supported application performance monitoring, troubleshooting, dashboarding, and production-readiness analysis across Dynatrace, CA APM, AppDynamics, and JMX-monitored enterprise applications.
Technologies: Dynatrace, CA APM, AppDynamics, JMX, Performance Testing, Troubleshooting